

Task 2: Smart Irrigation Simple System Simulation

1. Simulation Overview:

This simulation demonstrates a smart irrigation system using Internet of Things concepts for automated agricultural monitoring. A soil moisture sensor continuously measures soil conditions and sends analog data to the Arduino controller. Based on a predefined threshold value, the system automatically activates or deactivates irrigation. Visual indicators using red and green LEDs display the current soil condition, while a relay controls the pump motor connected to an external power supply.

2. Components Used:

- a. Arduino Uno
- b. Soil Moisture Sensor
- c. Breadboard
- d. Relay SPDT
- e. DC Motor
- f. Red LED
- g. Green LED
- h. Resistors
- i. External Power Supply

3. Simulation Screenshots:

Important Note: During simulation, the relay was controlled directly by the microcontroller output pin. In a practical hardware implementation, a transistor driver circuit would be recommended to reduce current load on the microcontroller and improve circuit reliability.

- Full Circuit Design:

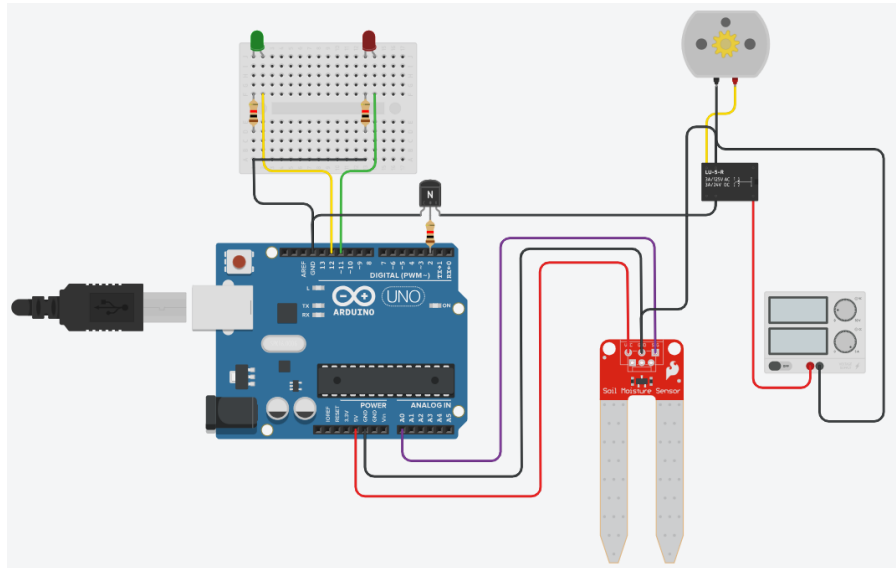


Figure 1: Complete Smart Irrigation Circuit Design

- Dry Soil Condition:

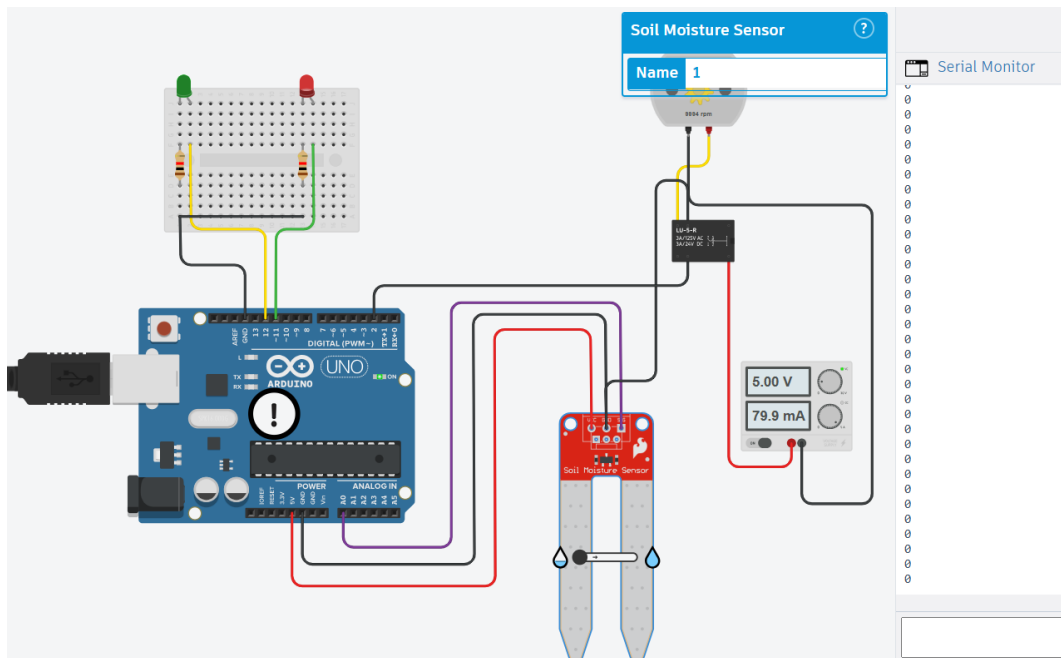


Figure 2: Dry Soil Detected (value=0<threshold 400) – Irrigation Activated

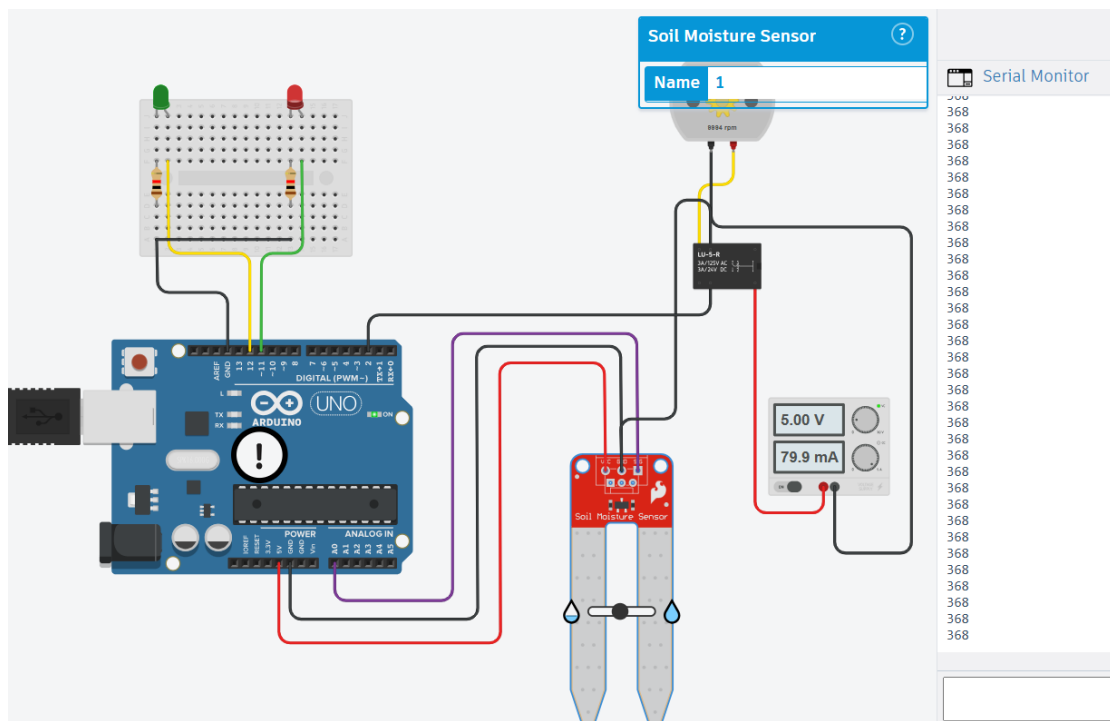


Figure 3: Dry Soil Detected ($0 < \text{value} = 368 < \text{threshold } 400$) – Irrigation Activated

- Wet Soil Condition:

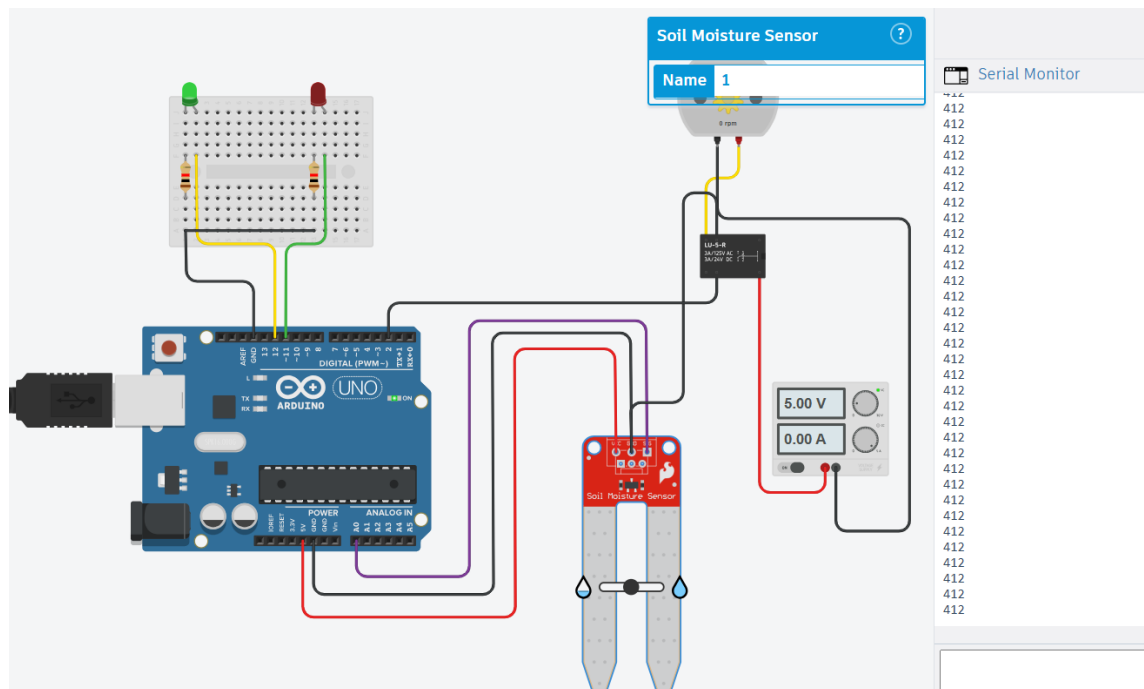


Figure 4: Wet Soil Detected ($\text{threshold} < \text{value} = 412 < \text{biggest value}$)- Irrigation Stopped

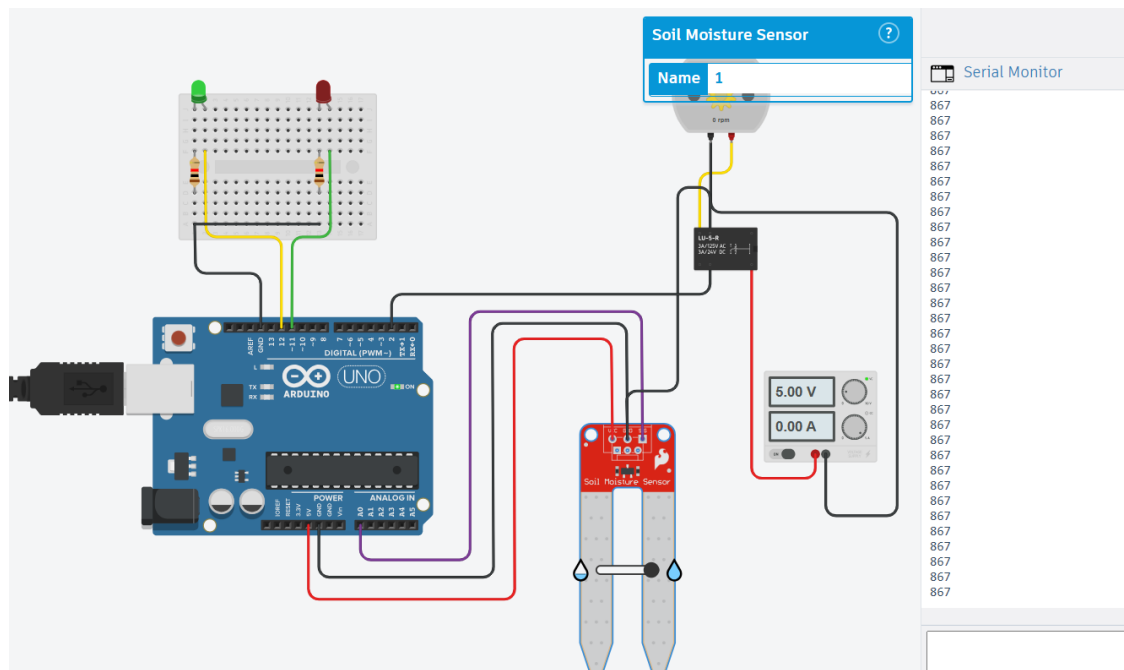
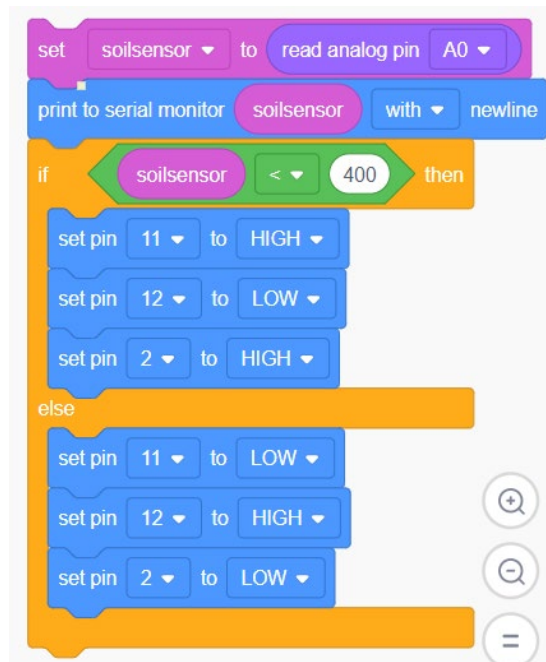


Figure 5: Wet Soil Detected (threshold< biggest value=867)-Irrigation Stopped

- Arduino Code:



4. C++ Code and Code Explanation:

```
int SoilSensorValue=0;
int redLED=11;
int greenLED=12;
int relayPIN=2;

void setup ()
{
  Serial.begin (9600);
  pinMode (redLED, OUTPUT);
  pinMode (greenLED, OUTPUT);
  pinMode (relayPIN, OUTPUT);
}

void loop ()
{
  SoilSensorValue = analogRead(A0);
  Serial.println(SoilSensorValue);

  if (SoilSensorValue < 400)
  {
    digitalWrite (redLED, HIGH);
    digitalWrite (greenLED, LOW);
    digitalWrite (relayPIN, HIGH);
  }
  else
  {
    digitalWrite (redLED, LOW);
    digitalWrite (greenLED, HIGH);
    digitalWrite (relayPIN, LOW);
  }
}
```

```
}  
  delay (10); // Delay a little bit to improve simulation performance  
}
```

The Arduino program controls an automated smart irrigation system based on soil moisture conditions.

First, the variable **SoilSensorValue** is declared to store the analog value read from the soil moisture sensor.

In the **setup** function, communication with the serial monitor is initialized using **Serial.begin (9600)**. The LED pins and relay pin are configured as output pins.

Inside the **loop ()** function, the Arduino continuously reads the soil moisture value from analog pin **A0** using **analogRead(A0)**. The sensor value is then displayed on the serial monitor for monitoring purposes.

An **IF-ELSE statement** compares the sensor value with a threshold value of 400. If the sensor value is below **400**, the system interprets this as dry soil. The Arduino turns on the red LED connected to pin 11 and activates the relay connected to pin 2, which powers the water pump motor.

If the sensor value is **400 or above**, the soil is considered sufficiently moist. The Arduino turns off the red LED and the relay, while activating the green LED connected to pin 12 to indicate normal soil conditions and moisture.

A short delay is added to improve simulation stability and prevent rapid switching.

